NN & CNN LAB PROGRAMS

Short Exam Versions

1. Artificial Neuron

```
import numpy as np
x = [2, 3]
w = [0.5, 0.5]
b = 1
y = np.dot(x, w) + b
print(y)
```

2. Sigmoid and ReLU Functions

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-5,5,100)
sig = 1/(1+np.exp(-x))
relu = np.maximum(0,x)
plt.plot(x,sig,label="Sigmoid")
plt.plot(x,relu,label="ReLU")
plt.legend()
plt.show()
```

3. Feedforward Neural Network – NumPy

```
import numpy as np
x = np.array([[0,0],[0,1],[1,0],[1,1]])
w = np.array([[1,-1],[1,-1]])
b = np.array([0,1])
z = np.dot(x,w) + b
y = 1/(1+np.exp(-z))
print(y)
```

4. Image Convolution

```
import cv2
import numpy as np
img = cv2.imread("image.jpg",0)
k = np.ones((3,3))/9
out = cv2.filter2D(img,-1,k)
cv2.imshow("Result",out)
cv2.waitKey(0)
```

5. Image Filters – Blur and Edge Detection

```
import cv2
img = cv2.imread("image.jpg")
blur = cv2.GaussianBlur(img,(5,5),0)
```

```
edge = cv2.Canny(img,100,200)
cv2.imshow("Blur",blur)
cv2.imshow("Edge",edge)
cv2.waitKey(0)
```

6. Max and Average Pooling – NumPy

```
import numpy as np
x = np.array([[1,2,3,4],
              [5,6,7,8],
              [9,10,11,12],
              [13,14,15,16]])
print("Max:", np.max(x.reshape(2,2,2,2),axis=(2,3)))
print("Avg:", np.mean(x.reshape(2,2,2,2),axis=(2,3)))
```

7. CNN Model – Keras

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Conv2D,MaxPooling2D,Flatten,Dense
m = Sequential([
    Conv2D(32,(3,3),activation='relu',input_shape=(64,64,3)),
    MaxPooling2D(2),
    Flatten(),
    Dense(64,activation='relu'),
    Dense(10,activation='softmax')
])
m.summary()
```

Required Libraries

Install all required libraries using this single command:

```
pip install numpy matplotlib opencv-python tensorflow
```

Note: Keep image.jpg in the same folder as the image programs.